

FACULTY ELECTRICAL ENGINEERING, MATHEMATICS
AND COMPUTER SCIENCE

Department of Microelectronics

[EE4605] Integrated Circuits for RF/Wireless Applications

Assignment 2
LNA Design

ANASTASIOS KAZAKOS
6398308

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1 Problem 1 [Device Characterization]

The operating frequency for this problem and for the following ones is 2.08 GHz. The setup of the simulation was given, and it is presented in figure (1.1) with the necessary connections.

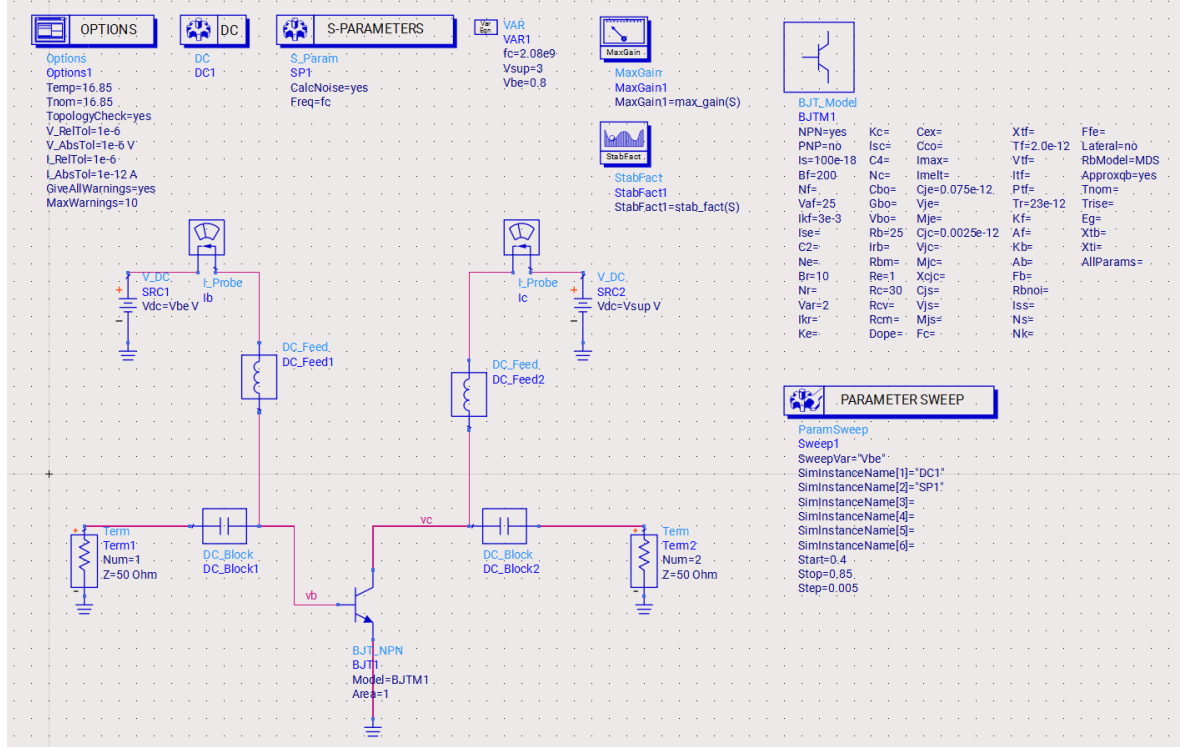


Figure 1.1: Device Characterization Setup.

1.1 Questions

a i)

Determine the f_T and F_{min} of the reference transistor using a V_{CE} of 3 V as a function of the collector current (use log scale).

For this simulation, I perform a parametric sweep of the bias voltage of the BJT transistor (V_{be}) for the given operating frequency of 2.08GHz, as it is presented in figure (1.1). Afterwards, I plot the f_t and F_{min} with respect to the collector current in order to find the frequency for the maximum speed and the frequency for the lowest noise (figure 1.2).

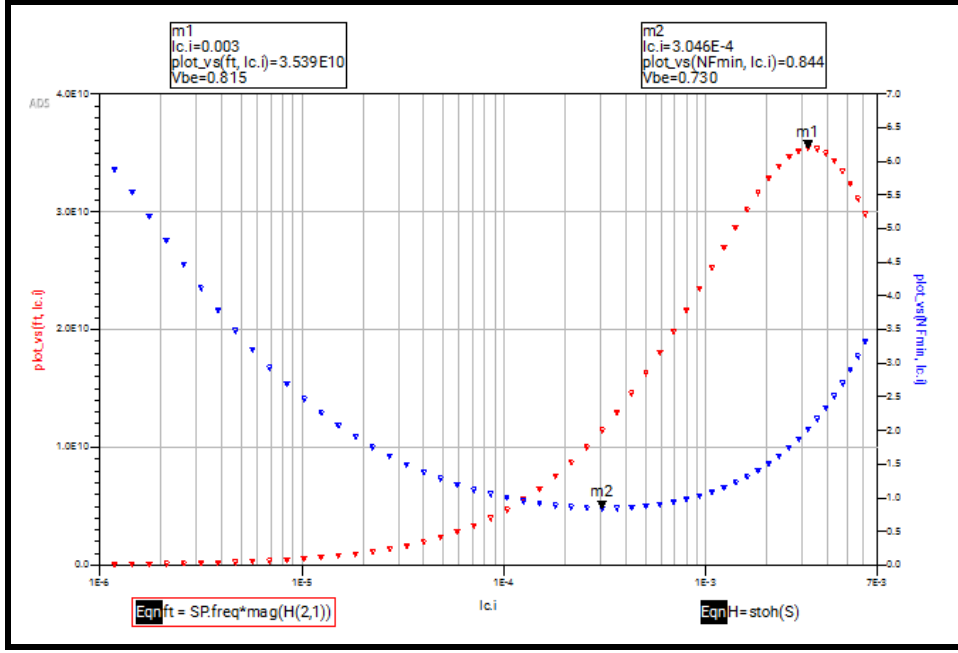


Figure 1.2: f_t and NF_{min} with respect to the collector current.

As it is illustrated in figure (1.2) the f_t and F_{min} are:

$f_t = 35.39GHz$	$F_{min} = 0.844$
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a ii) Explain the choice of your simulation frequency.

The way we find the f_t was through extrapolation, assuming that we have an one pole device. This means that the current gain of the device until it reaches unity current gain (at f_t) has only one pole. This assumption allow us to choose any frequency after the -3dB frequency and, afterwards, calculate the cut-off frequency through extrapolation using the h-parameters. In particular, we use the h(2,1) parameter since it gives us the ratio of collector current over base current. The equation used to calculate the f_t is presented below. Moreover, it is clearly depicted in figure (1.3) that the operation frequency is after the -3dB frequency so the extrapolation technique can be used. However, I have to state that even this technique doesn't take into account the zero which can be close to the unity current gain frequency causing some error in the calculation of f_t .

$$f_t = f_{measured} * |h(2, 1)| \quad (1.1)$$

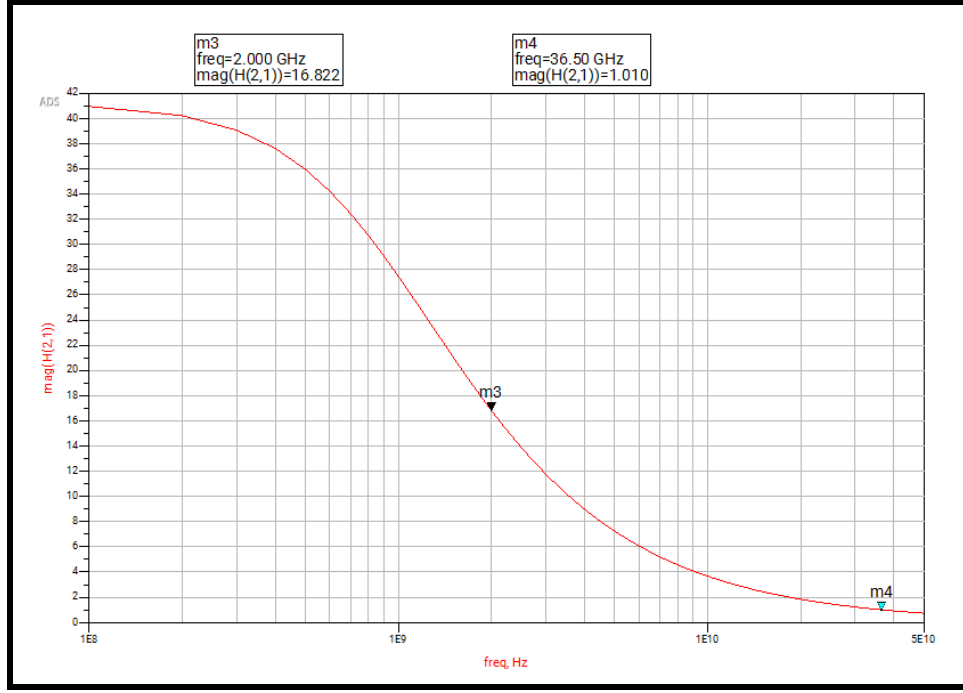


Figure 1.3: Proof that the operation frequency is at the one pole slope (the scale of the gain is not in dB).

a iii) What is the best biasing point for low noise?

As it is observed, for $V_{BE} = 0.730V$ the minimum noise figure is achieved. Moreover, I have to state that this minimum noise figure is when we apply the optimum noise matching, meaning that biasing the base at this voltage does not guarantee the minimum noise figure.

b) Is this also the point for achieving high transistor gain?

The point where we achieve high transistor gain is when we bias for the maximum f_t and not for the minimum noise figure. Having a large f_t moves the one pole slope of the gain in higher frequencies. That means that the frequency spectrum where the gain remains constant is also getting bigger, making our amplifier more wideband. Moreover, the current flowing through the transistor near the f_t is big, further increasing the gain of the amplifier. Biasing our device after f_t is usually not suitable, since we lose in bandwidth, and we begin to have distortion issues. High power consumption and higher noise are also some reasons which make the biasing in this region not appealing, at least for our design. In conclusion, when we aim for high transistor gain, we have to bias our device for maximum f_t .

c) Plot the K-factor as a function of frequency for the optimum bias point, Is the device unconditionally stable?

Since we design an LNA we aim for minimum noise figure, meaning that we have to bias our device at $V_{BE} = 0.730V$. Afterwards, I sweep the frequency to investigate if our device is stable not only in the operation frequency (or close to it), but for all the frequencies that could potentially exist for our device. At lower frequencies, the gain is high, meaning that this could lead to unstable behaviours. If the system is

oscillating at another frequency bandwidth, then this will ruin the performance. The plot of the K-factor is presented in figure (1.4).

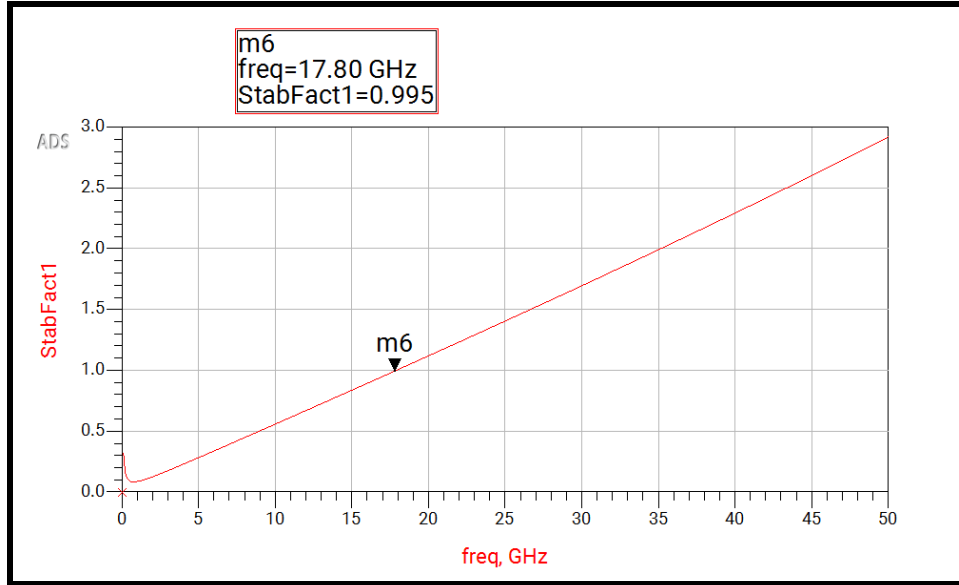


Figure 1.4: K-factor over frequency.

From figure (1.4), we can observe that the device is conditionally stable, since below 18 GHz approximately it becomes unstable. In most cases, the devices are not unconditionally stable, meaning that a behaviour like this was expected.

2 Problem 2 [An inductively degenerated LNA stage]

The deliverable of this exercise is to perform noise and impedance matching, by finding the optimal combinations of component values, device scale, and collector current in order to meet the specifications below:

F	<1.2 dB
S_{11}	<-15dB
S_{22}	<-15dB
S_{21}	>18dB
K	>1

2.1 Choosing the biasing point

To begin with, I have to choose the biasing of the BJT device. Since we design an LNA, we aim for limited noise, while at the same time we try to minimize the power consumption and generate sufficient gain. Following these guidelines, I choose to bias my transistor to the point where I achieve a noise figure below 1.2dB with the smallest collector-emitter current possible in order to minimize the power consumption. The chosen biasing point is presented below **2.1**:

$$V_{be} = 0.75V \quad F_{min} = 0.920$$

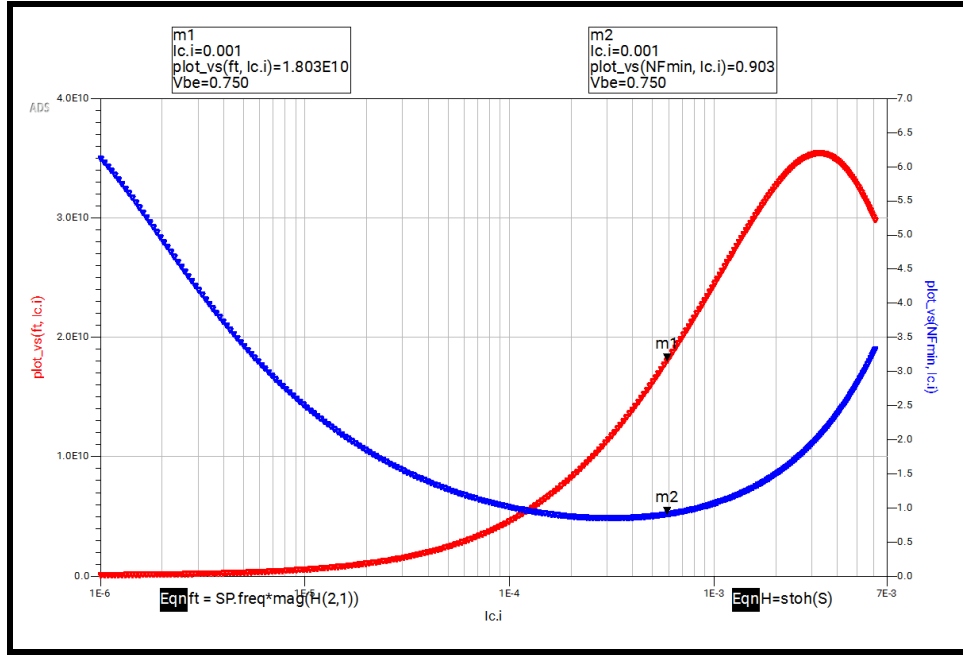


Figure 2.1: Biasing point.

We can see in figure (2.2) that for the operating frequency, the noise factor is way above the requirement (1.2dB) and the forward transmission coefficient ($|S_{21}|$) of the transistor (raw gain of the device) is way below the wanted value of at least 18dB. However, the Maxgain and the minimum noise factor indicates that there is the potential of achieving the requirements. For this reason, we have to perform two tasks, noise matching to reduce the noise factor and impedance matching to boost the gain of the device. Finally, I have to state that since we aim for low noise, probably we won't achieve the maximum gain illustrated in figure 2.2.

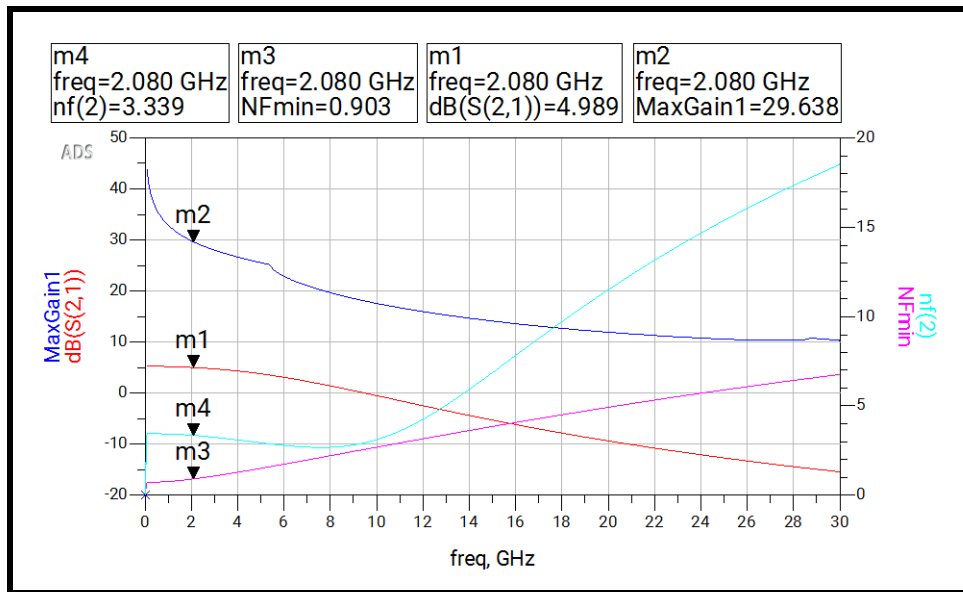


Figure 2.2: Forward Transmission Coefficient and maximum gain.

2.2 Noise and input impedance matching

During the matching, we need to keep the current density constant in order to keep the noise figure and the gain the same. If the current density becomes smaller then the gain falls and this way the signal-to-noise ratio becomes smaller (the noise factor increases). If the current density becomes too high, it starts to drop due to high current effects (less gain) and the noise factor becomes even higher. To keep the current density constant we have to increase the current with the same manner as we increase the area (as we see from equation 2.1). This is automatically done from the model.

$$I = J * A \quad (2.1)$$

Since the generator impedance is 50Ω , we want to move the input reflection coefficient and the optimum reflection coefficient for noise to the center of the smith chart for the operating frequency. Equations 2.2 and 2.3 represent the reflection coefficient for the noise and the optimum admittance for minimum noise respectively:

$$S_{opt} = \frac{Y_o - Y_{opt}}{Y_o + Y_{opt}} \quad (2.2)$$

$$Y_{opt} \approx gm(\frac{1}{\sqrt{\beta}} - j\frac{\omega}{\omega_T}) \quad (2.3)$$

From equation 2.2, we can observe that in order to move the optimum noise reflection coefficient to the origin we need to increase the admittance. By increasing the area of the device, and its current as well, the transconductance increase, and as a result the optimum admittance increase moving the point closer to the origin. On the other hand, the input reflection coefficient can be written:

$$S_{11} = \frac{Z_{in} - Z_o}{Z_{in} + Z_o} \quad (2.4)$$

We can observe that in order to achieve input impedance matching, we have to decrease the input impedance. For this case, I calculate the input impedance, by performing the Kirchhoff's law in the following schematic **(2.3)**. I have applied a voltage at the input and grounded the output. Afterwards, I will add the base inductor with just a summation of impedances, since it is in series.

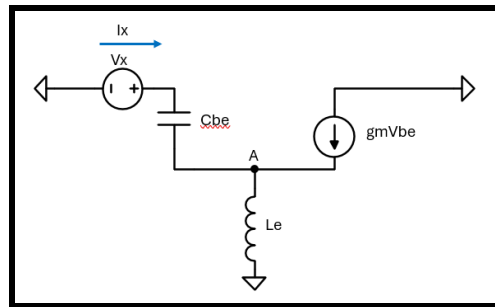


Figure 2.3: Small signal model of the BJT with the emitter inductor.

By performing KCL at node A:

$$V_x j\omega C_{be} + gmV_{be} = V_A \left(\frac{1}{j\omega L_e} + j\omega C_{be} \right)$$

For the input current:

$$I_x = (V_x - V_A) * j\omega C_{be} \Rightarrow V_A = -\frac{I_x}{j\omega C_{be}} + V_x$$

For the base emitter voltage:

$$V_x = V_{be} + V_A$$

By combining the above equations and adding the impedance base inductor:

$$Z_{in} = j\omega L_e + j\omega L_b + \frac{gmL_e}{C_{be}} + \frac{1}{j\omega C_{be}} \quad (2.5)$$

Right now, we can begin our matching. At first, we decrease the optimum impedance for noise matching Z_{opt} , by increasing the device area (I remind here that I want to increase the optimum admittance). I want to achieve a real impedance of 50Ω (or 1 for normalized impedances).

Afterwards, I will decrease the emitter inductor in order to make the real part of the input impedance equal to 1 as well, since it affects the real part of the input impedance (equation 2.5).

Finally, I increase the base inductor for two reason. In the case of the input impedance, a larger base inductor will increase the imaginary part, which is what it is needed for impedance matching. In the case of the optimum noise impedance, we know that the nature of the impedance at the input of a BJT transistor is capacitive. For this reason we need an inductor at the input to cancel out this behaviour and achieve noise matching. This can also be seen from equation 2.3 from the minus sign of the expression.

In the figures **(2.4)**, the noise figure and the output reflection coefficient are illustrated, while in figure **(2.5)** we can observe that the input impedance and noise matching has been achieved adequately and the current that is being consumed is 4.622mA. The values of L_e , L_b , and the scaling factor are:

$$L_e = 0.308nH \quad L_b = 2.378nH \quad Area = 7.85$$

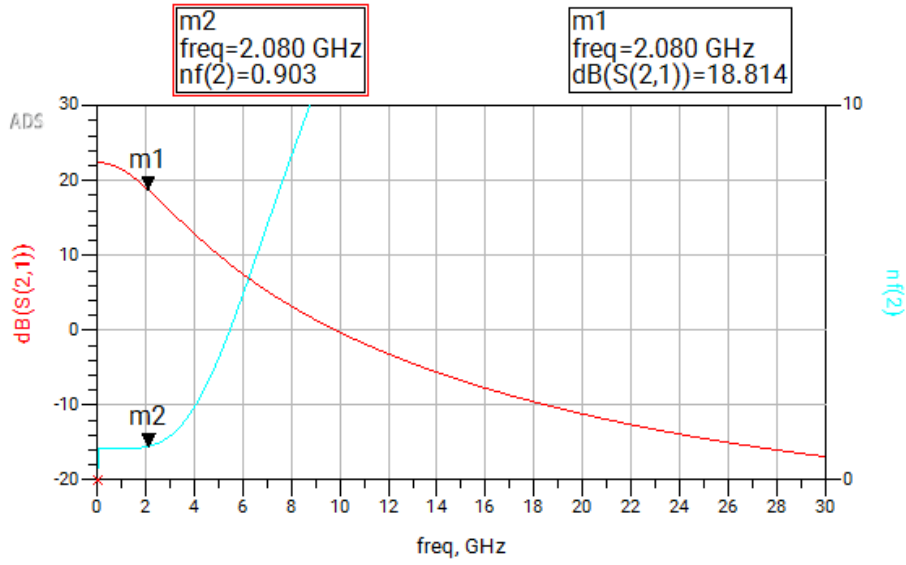


Figure 2.4: Test bench of the device, along with the values of the area, base inductor and emitter inductor.

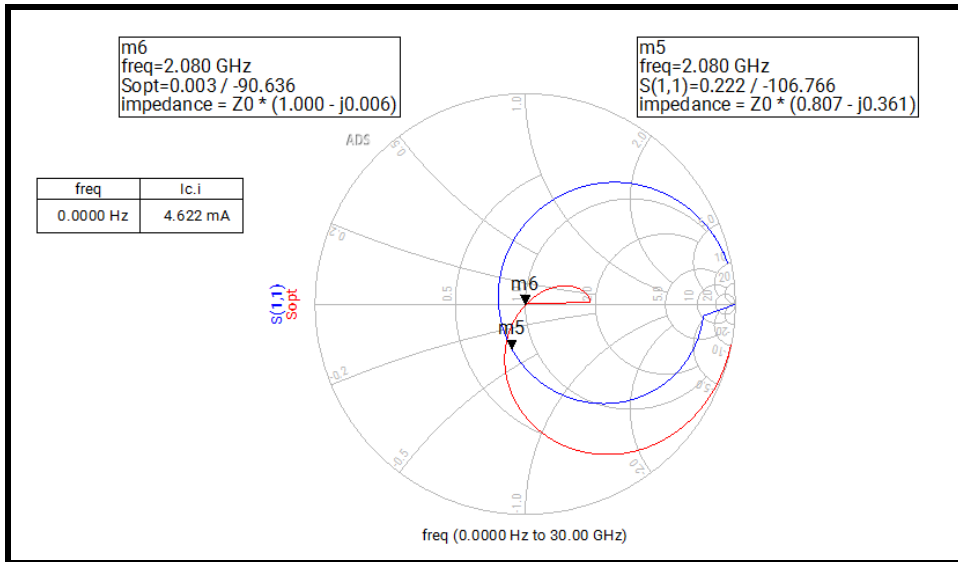


Figure 2.5: Impedance and noise matching, along with the current value.

Even though the gain could become a bit smaller by decreasing V_{be} , we know that afterwards we will proceed with the stabilization of the device. This process will decrease our gain for reasons that I will explain later.

2.3 Stability

Before choosing the technique that we will use to make the device stable, I will check where it becomes unstable. From figure (2.6), we can observe that nearly at all frequencies the circuit is unstable. That, means that we have to reduce the gain in all frequencies. This leads us to the conclusion that we need to use a single resistor in order to reduce the gain in all frequencies. The resistor should be placed at the output of our device, since we are designing an LNA and we prioritize for low

noise. If we had put it in the input, then the noise of the resistor will be amplified, raising to a great extent the noise figure. I choose to put a 500Ω shunt resistance at the output. This way some of the output signal will be drawn to ground reducing our gain, and lowering the risk of oscillation. This tactic will guarantee a stability factor over 2, approximately, for all frequencies (figure 2.7). We can also see that the gain dropped as it was expected from 18.814dB to 18.015dB, a difference of 0.799dB.

In practice, we wont operate the device after ft, but I plot the above quantities to a bigger range.

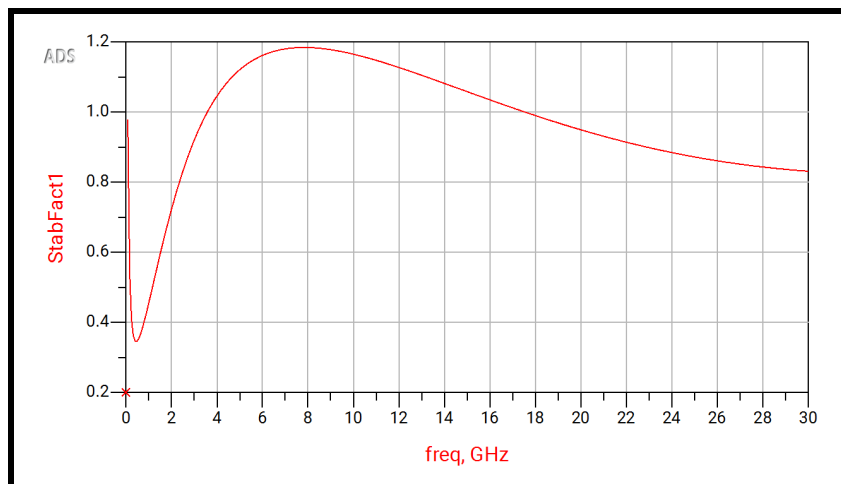


Figure 2.6: Unstable behaviour for the majority of frequencies.

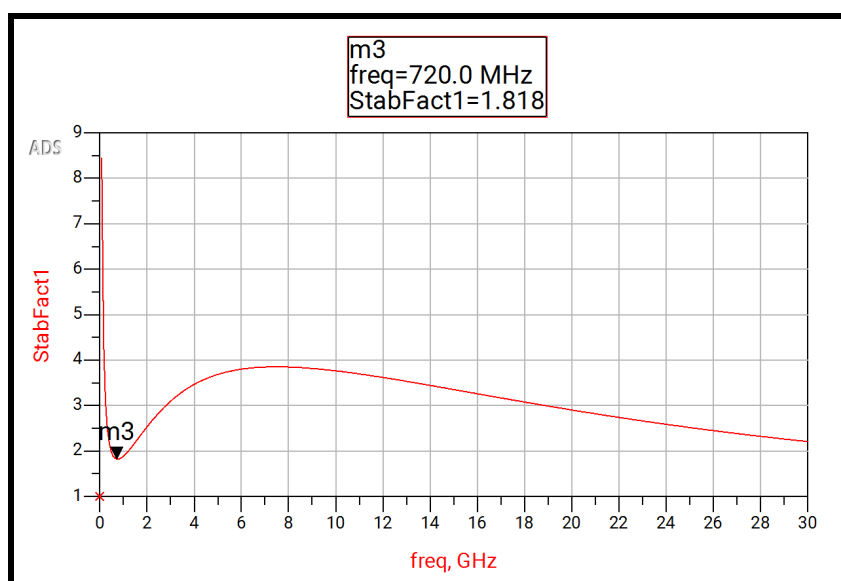


Figure 2.7: Stability achieved after killing some gain with a shunt resistor at the output.

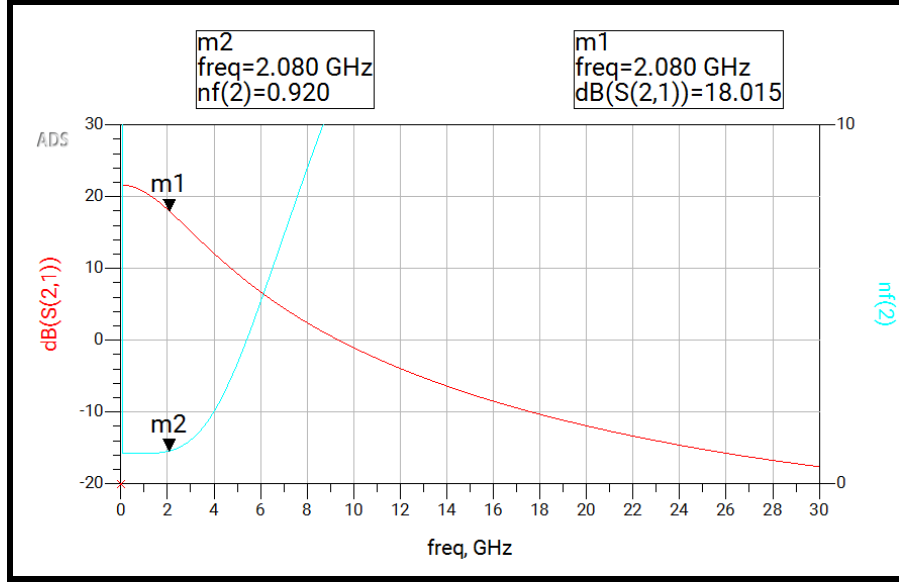


Figure 2.8: Reduction of the gain.

2.4 Output matching

We already achieved for the input reflection coefficient to be close to zero. We need to perform output matching, in order to make the reverse transmission coefficient to go to zero as well. Achieving this value for the s-coefficients, or at least be close, we minimize the losses due to the reflection of our signals at the input and the output. In other words, by performing good matching we earn in low power losses and low interference as well, because the reflected signal interferes with the transmitted signal. In figure (2.9) the location of S_{22} on the smith chart at the operating frequency can be seen. Since it is located on the down right half, I choose to use a shunt capacitor to move downwards in the Y-smith chart and reach the unity impedance circle. Afterwards, I use a series inductor to move upwards along the unity impedance circle and reach the origin. The result after matching for the operating frequency are presented in figure (2.10).

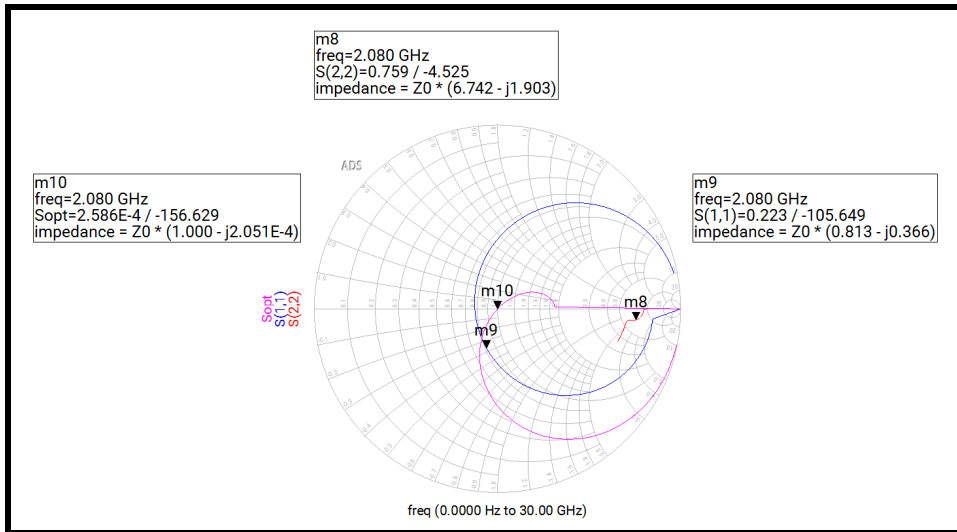


Figure 2.9: Point of S_{22} for the operating frequency before matching.

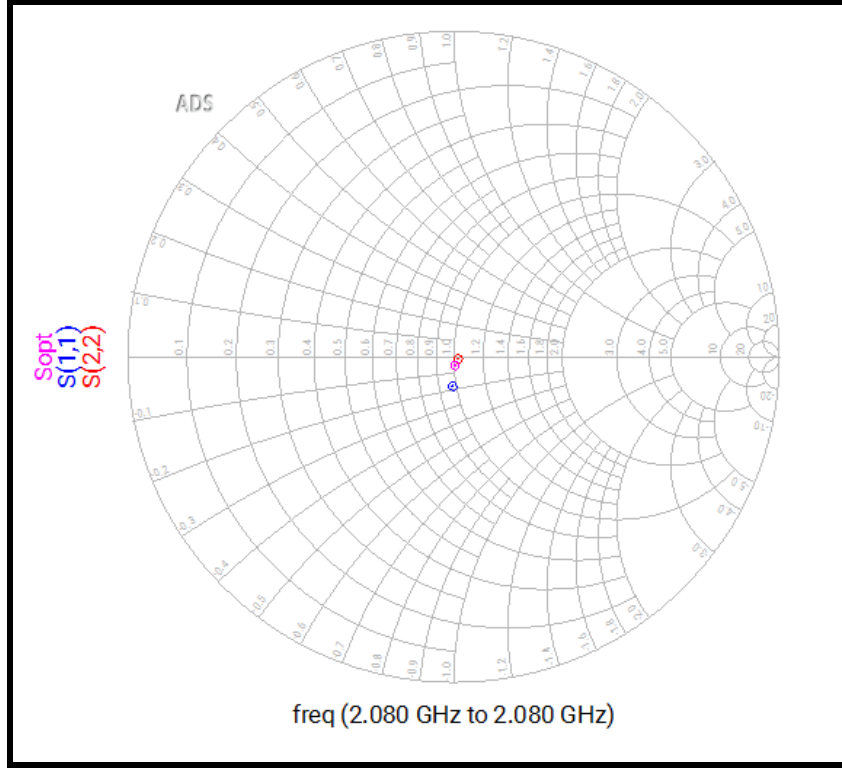


Figure 2.10: Output matching.

2.5 Tuning for optimum performance

After the output matching the gain has been raised, allowing us to further decrease the biasing point V_{be} and improve the power consumption. Moreover, the stability factor has been improved because at high frequencies there is another path that goes to ground (the inductor), adding to the stabilization of the circuit. Finally, the input reflection coefficient experienced some changes as well. As we can see from figure 2.3 and the following equation, the input voltage is dependent from the voltage at node A. Any slight impedance change at the output will alter the voltage at node A and as a result the input voltage, changing the amplitude of the waves at the input and thus changing the input reflection coefficient as well. We can say that this inductor provides feedback to the input. It seems that the output matching has also a small impact in noise figure, which is logical. The inductor L_e has an internal resistance, which means that changing the voltage at node A will also alter the noise performance. The same can be assumed for the inductor at the base. After make the necessary tuning, the resulted S_{21} , MaxGain, stability factor, current drawing and matching is presented below. The values of the parameters are also presented.

$$V_{be} = 0.729V \quad Area = 7.68 \quad L_e = 0.5576nH \quad L_b = 3.638nH \quad R = 500\Omega$$

$$C1 = 0.455pF \quad L3 = 10.222nH \quad I_{ce} = 2.26mA$$

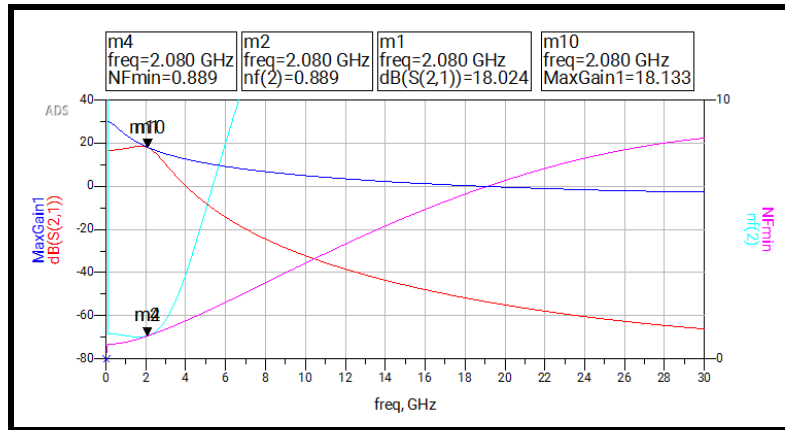


Figure 2.11: S₂₁, MaxGain, noise figure and minimum noise figure.

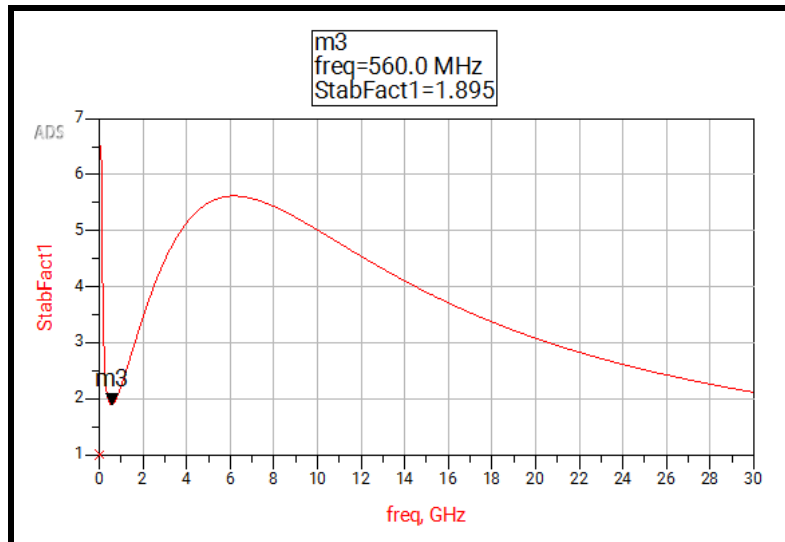


Figure 2.12: Stability factor.

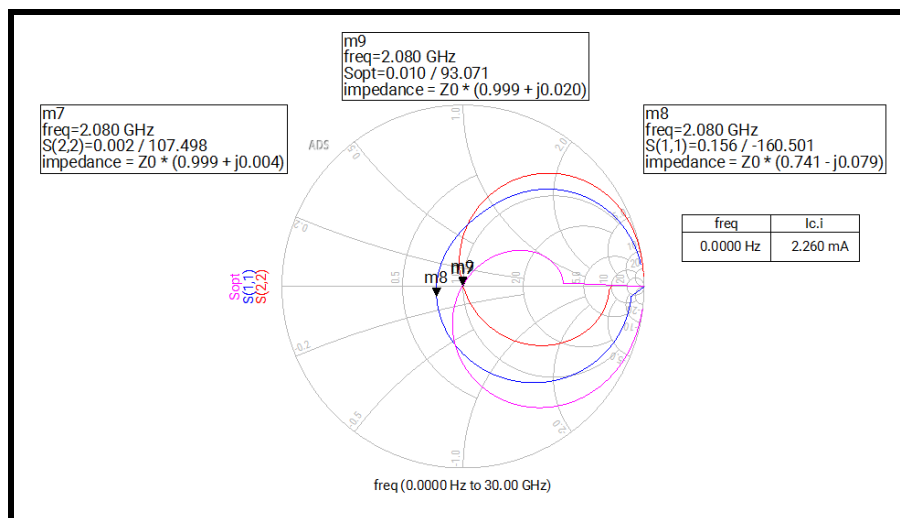


Figure 2.13: Matching and current drawing.

From figure (2.11), we can observe that S_{21} is close but not equal with the Max possible Gain. This is logical since in order to achieve the smallest possible noise figure we had to sacrifice in gain. Our matching is made with low noise as a goal and not maximum gain. Finally in figure (2.14) I plot the s-parameters, where the requirements for the operating frequency have been met.

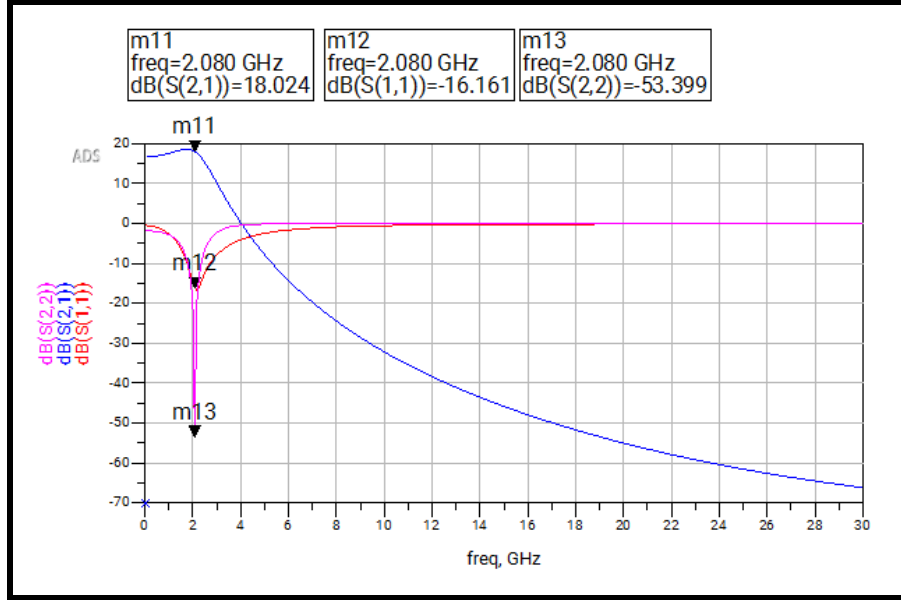


Figure 2.14: S-parameters.

2.6 Unanswered question until now

1) Why does the emitter inductance has much more impact on the input impedance than the noise matching condition?

The emitter inductance as we can see from equation 2.5 creates a noiseless imaginary and real part to the input impedance. That clearly explain why it is mainly affects the impedance matching. While it also affects the noise matching since in reality the inductor has an intrinsic resistance as well, the effect in noise is small. The reason is that the optimum admittance for noise is mainly affected by the shot noise of the BJT device and the thermal noise at the base. That's the reason why the base inductor affect in a greater extent the noise performance.

2) How does the F of the LNA compare to the Fmin of the device?

The noise figure can become equal with the minimum noise figure only in the case of optimum noise matching. This can be seen also from the expression of the noise figure:

$$F = F_{min} + \frac{R_n}{G_g} [(G_{opt} - G_g)^2 + (B_{opt} - B_g)^2] \quad (2.6)$$

As we can observe only in the case where $G_{opt}=G_g$ and $B_{opt}=B_g$ the noise figure can become equal to F_{min} .

3) Why is inductive degeneration better for noise than resistive degeneration?

As discussed in question 1, the inductor at the emitter creates a noiseless real and imaginary impedance part at the input. If we used a resistance instead of the inductor, then it would add a noisy part at the input, deteriorating the noise performance.

3 Problem 3

The logic that I follow was to try to increase the gain of the circuit using a different topology in order to be able to reduce the base-emitter voltage and reduce as a result the current. I thought of using two CE stages, one on top of the other. As it is shown in figure (3.1) the CE stage at the bottom remained the same (Q1). The emitter of the CE stage on top (Q2) is connected with the collector of the transistor (Q1) through an inductor in order to pass only the DC current. The AC signal at the collector of Q1 is transferred to the base of Q2 where it is amplified again. I bias Q2 at the supply voltage and also AC grounded its emitter, since I use it as a CE stage.

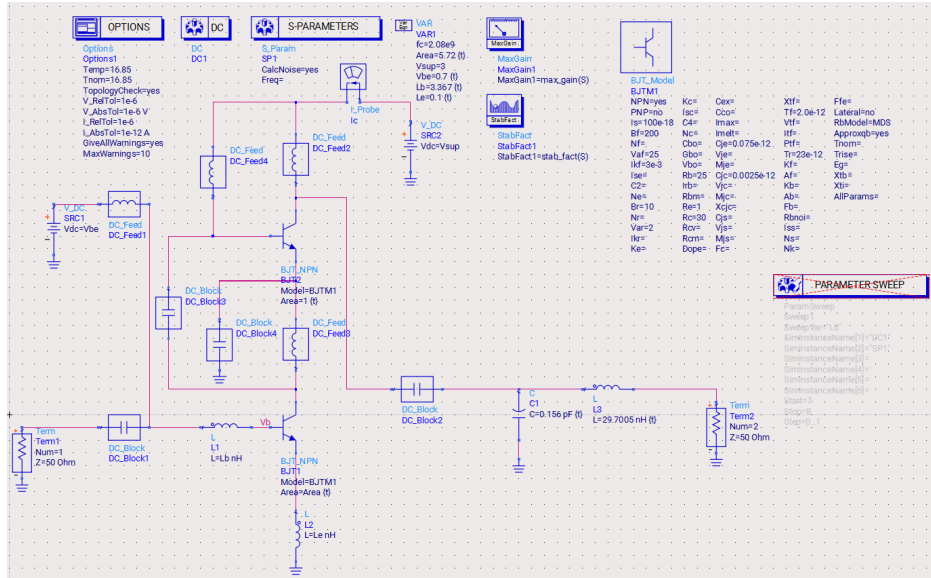


Figure 3.1: Schematic.

If we are willing to sacrifice 0.130dB of noise figure, then this topology can save more than 1/4 of current. However, if we are unwilling to do this compromise, this topology can still save us more than 1/2 of current. I would argue that the advantages of this topology is that it is unconditionally stable without a need of a stabilization network and the gain is more than double of the previous topology. However, a lot of inductors and capacitors was used for this design which will make the layout design tough.

Below both approaches are presented, the one with the noise figure almost equal with the previous one (difference of 0.03dB biased at 0.71V) and the one with the lower current consumption (biased at 0.7V).

3.1 Biasing at 0.71V

The figures below show:

Figure 3.2, the noise figure, F_{min} , S_{21} and the MaxGain.

Figure 3.3, the noise matching, the input and output impedance matching along with the current consumption.

Figure 3.4, the remaining s-parameters.

The values of the parameters are:

$$V_{be} = 0.729V \quad Area = 7.68 \quad L_e = 0.5576nH \quad L_b = 3.638nH \quad R = 500\Omega$$

$$C1 = 0.455pF \quad L3 = 10.222nH \quad I_{ce} = 2.26mA$$

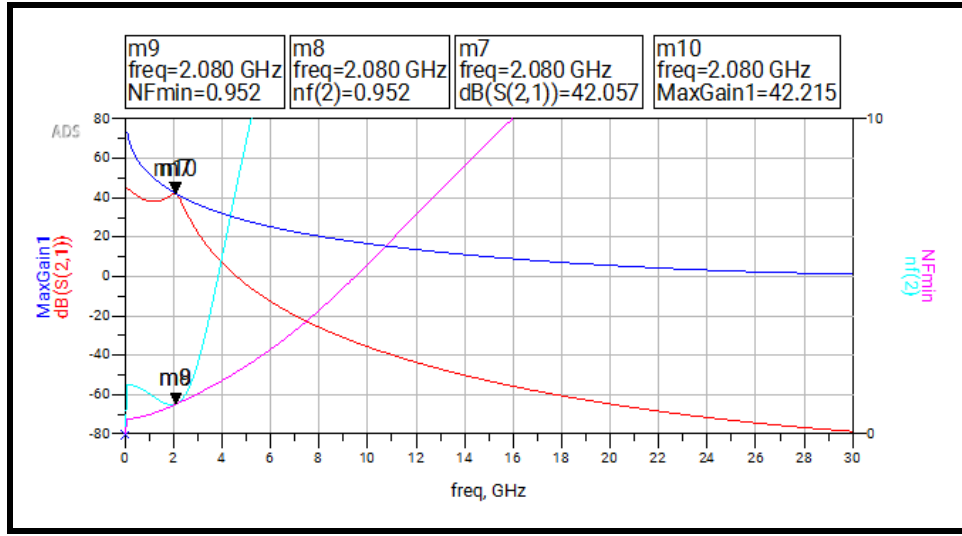


Figure 3.2: Noise figure, F_{min} , S_{21} and MaxGain.

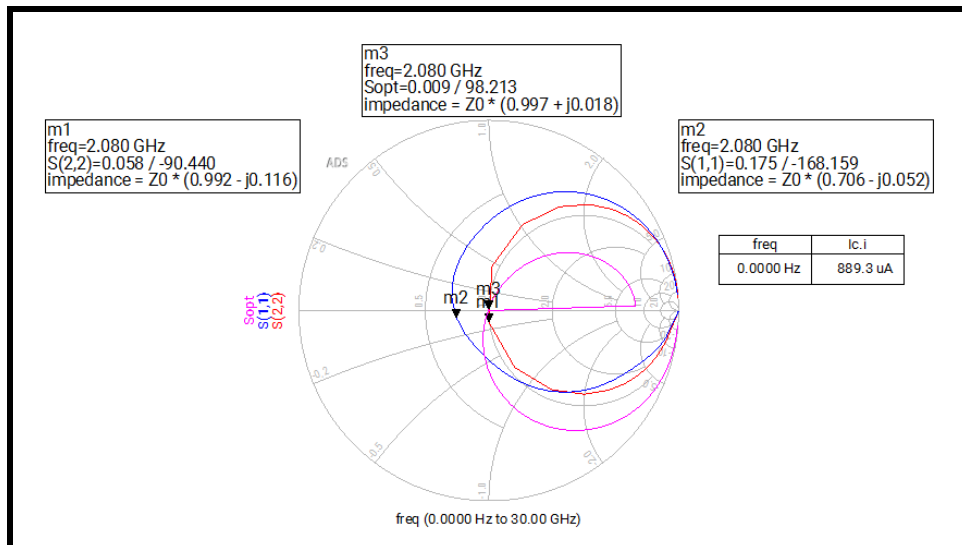


Figure 3.3: Noise matching, input and output impedance matching.

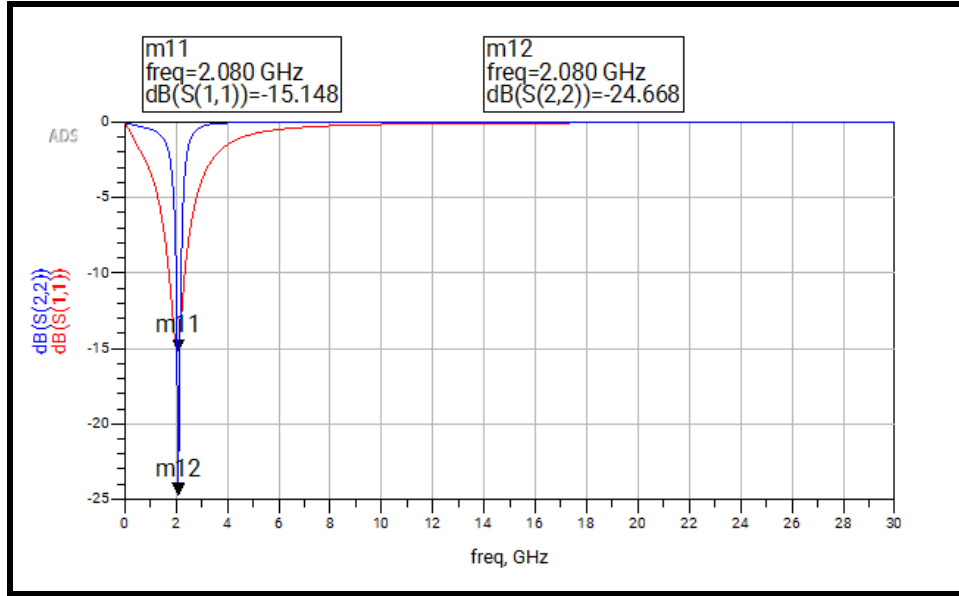


Figure 3.4: S-parameters.

3.2 Biasing at 0.7V

The figures below show:

Figure 3.5, the noise figure, F_{min} , S_{21} and the MaxGain.

Figure 3.6, the noise matching, the input and output impedance matching along with the current consumption.

Figure 3.7, the remaining s-parameters.

$$V_{be} = 0.71V \quad Area = 6.21 \quad L_e = 0.564nH \quad L_b = 5.914nH$$

$$C1 = 0.184pF \quad L3 = 29.701nH \quad I_{ce} = 889.3\mu A$$

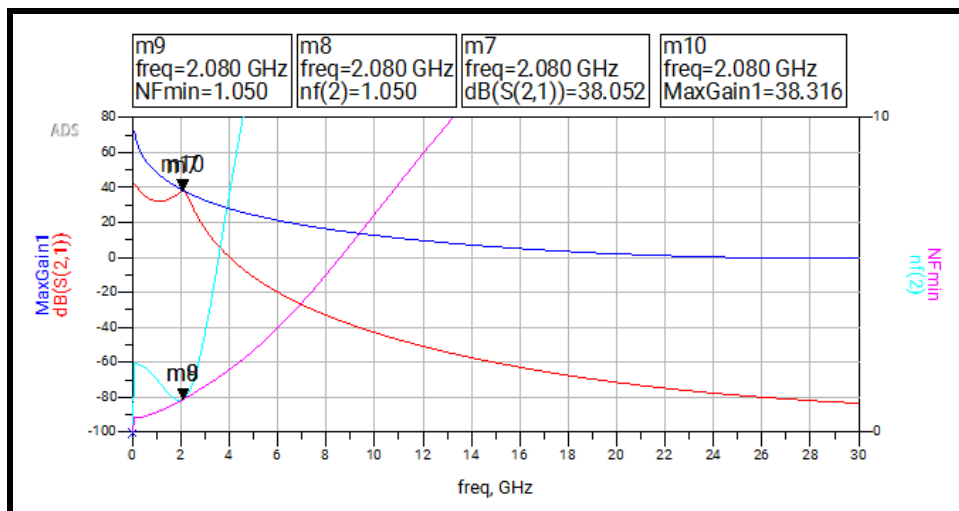


Figure 3.5: Noise figure, F_{min} , S_{21} and MaxGain.

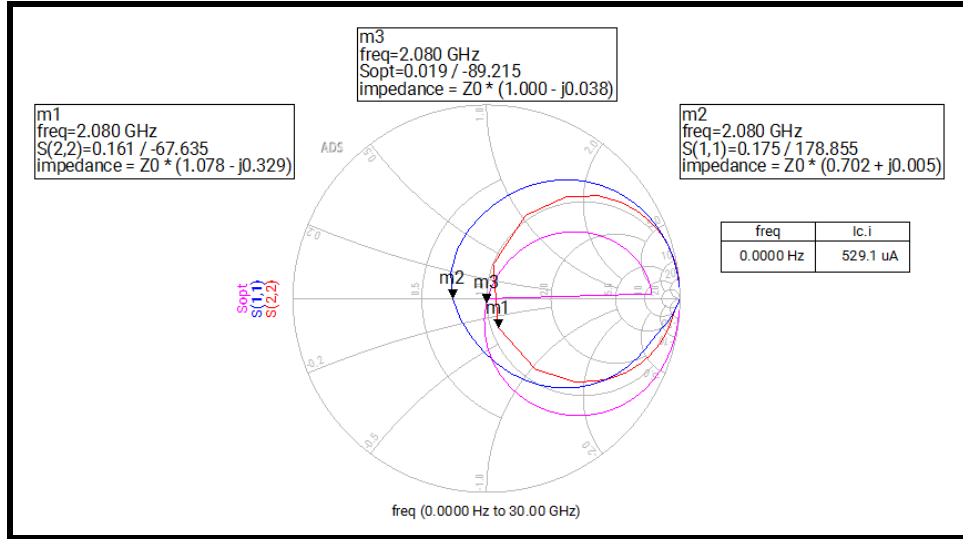


Figure 3.6: Noise matching, input and output impedance matching.

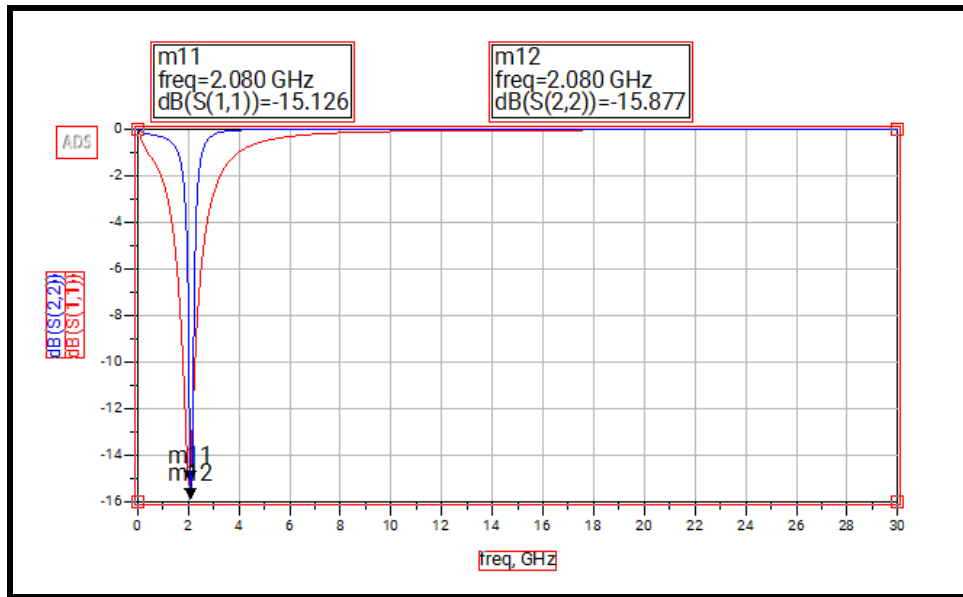


Figure 3.7: S-parameters.

$$V_{be} = 0.7V \quad Area = 5.345 \quad L_e = 0.784nH \quad L_b = 7.454nH$$

$$C1 = 0.154pF \quad L3 = 35.11nH \quad I_{ce} = 529.1uA$$

3.3 Stability

As it was stated above, in both cases the circuit is unconditionally stable. Even though the device will not operate after ft, I plot stability in a larger spectrum.

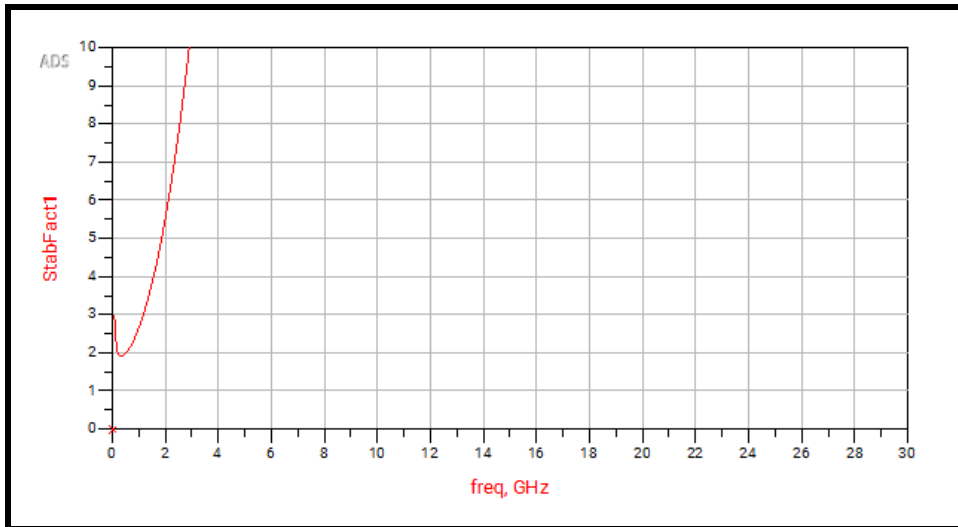


Figure 3.8: Stability factor for the first case ($V_{be} = 0.71$).

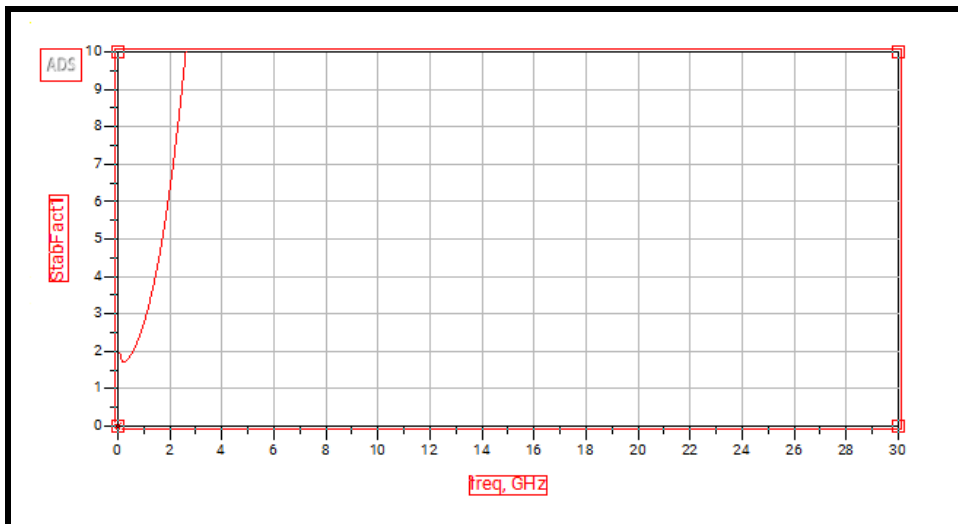


Figure 3.9: Stability factor for the first case ($V_{be} = 0.7$).